

# Bidirectional Mass Spectrometry Modeling Aided by Auxiliary Intermediate Steps From Graph Transformation Tools

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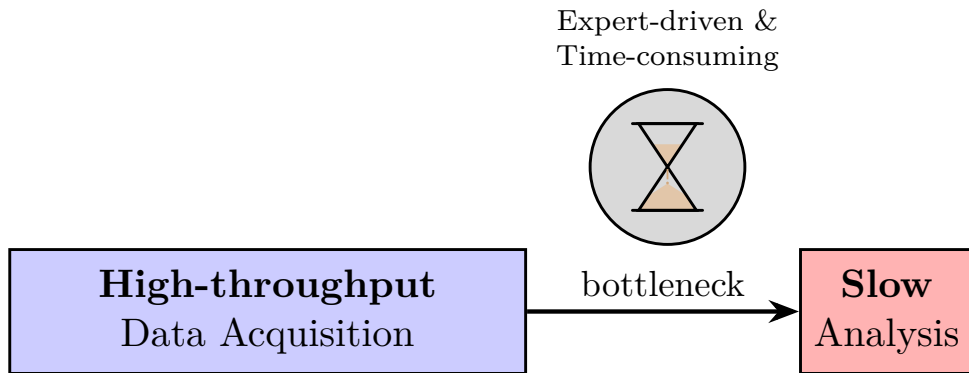
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2026-08-27

# Outline

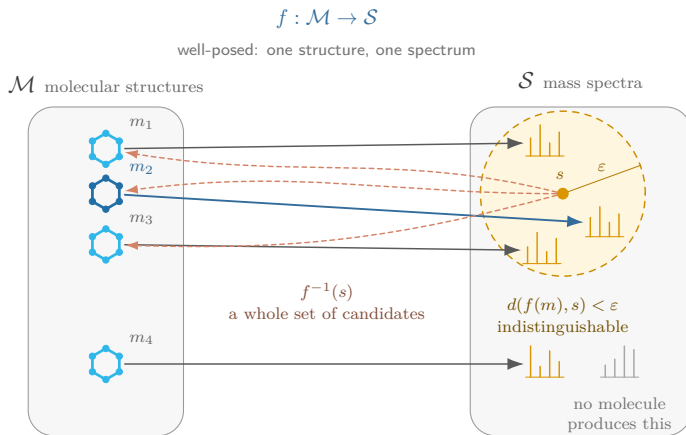
- 1 Problem
- 2 Background
- 3 Data
- 4 ML-Modeling
- 5 Results
- 6 Conclusions

# Motivation



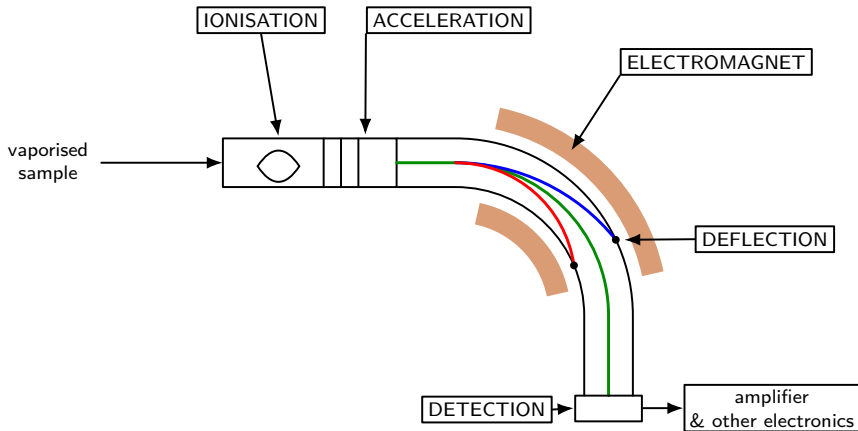
- Mass spectrometry is widely used, but analysis is the challenge
- Data acquisition is fast, interpretation slow  $\Rightarrow$  the bottleneck

# Problem Statement



Structure elucidation is the inverse of a map that is not invertible.

# Mass Spectrometer

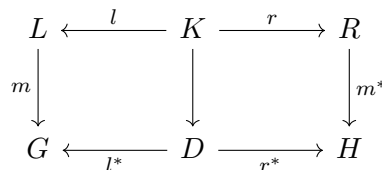


# Modelling Mass Spectra – An Open Problem

- **Rule-based** but limited to rule set
- **Data-driven** models do not generalize
- **Ab initio** does not scale
- currently <20% of detected compounds are identified

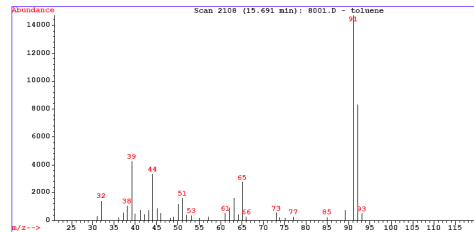
# MØD – A Graph Transformation Tool

- Molecules as labelled graphs, reactions as *rewrite rules*
- A rule is a span  $L \leftarrow K \rightarrow R$ :
  - $L$  – what must be present
  - $K$  – what is preserved (context)
  - $R$  – what it becomes
- Applied by *double pushout* (DPO) rewriting along a match  $m : L \rightarrow G$
- Repeated application builds the *derivation graph* of a reaction network
- Python interface `mod`: rules, strategies, predicates

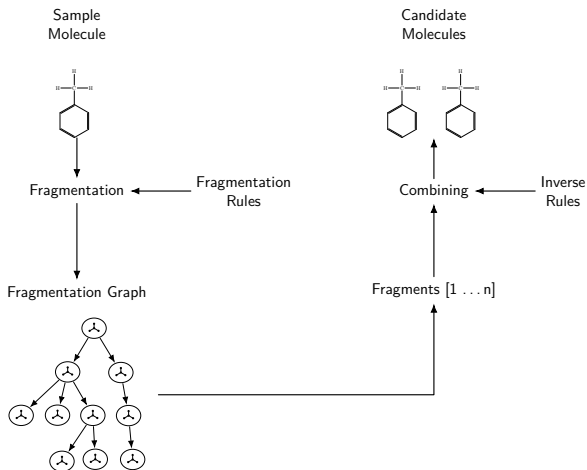


Host graph  $G \rightarrow$  product graph  $H$

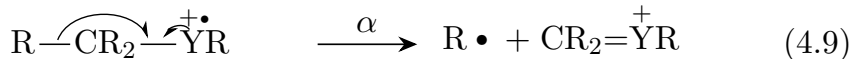
- Mass spectrometry data from NIST → SMILES strings & spectra
- Curated EI fragmentation reactions → encoded Graph Transformations



# Data Generation Overview



# Sample Fragmentation Rules

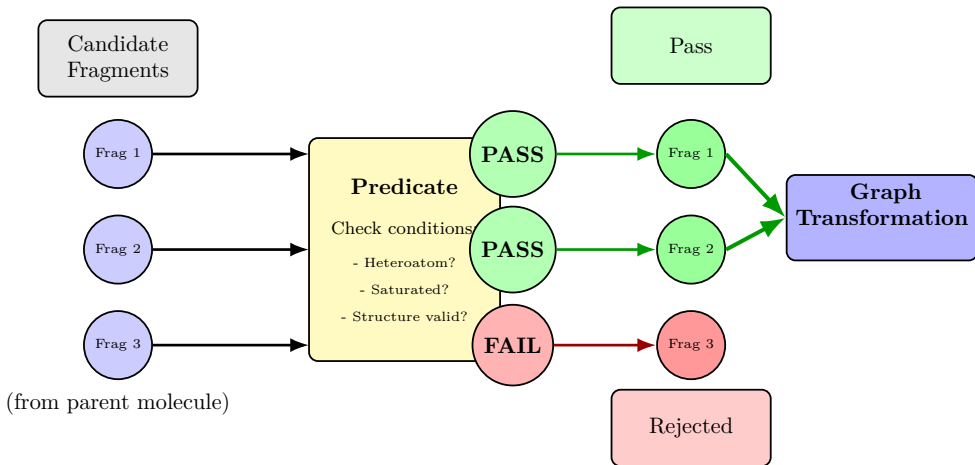


```

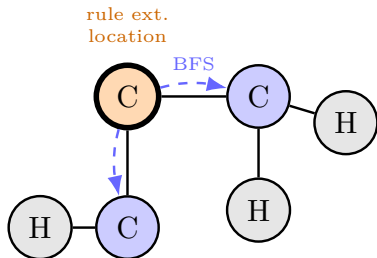
1      IMS_4_9 = mod.Rule.fromDFS(
2          s =
3              "[C]1[C]2([C]3)([C]4)[_A+.]5"
4              ">>"
5              "[C.]1" " ." "[C]2([C]3)([C]4){=}[_A+]5",
6          name =
7              "radical induced (alpha-)cleavage for a saturated site"
8              " 4.9"
9              " SR1R3R4Y5"
10     )

```

# Predicate

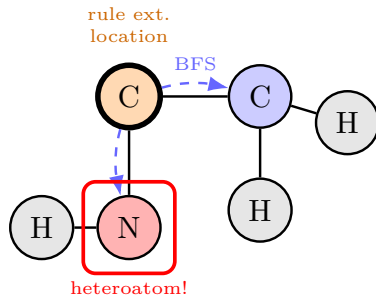


# Predicate – Alkyl Group



**VALID: Alkyl Group**

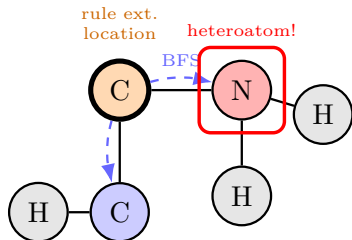
BFS neighbors  $\subseteq \{H, C\} \checkmark$



**INVALID: Not Alkyl**

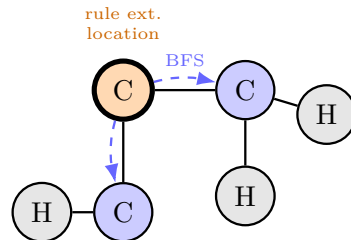
BFS neighbors  $\not\subseteq \{H, C\} \times$

# Predicate – Heteroatom Group



**VALID: Heteroatom Group**

$$\text{BFS} \setminus \{\text{H}, \text{C}\} \neq \emptyset \checkmark$$

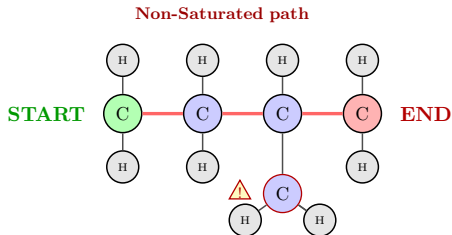
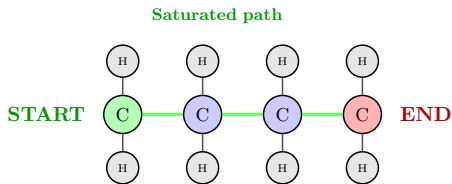


**INVALID: No Heteroatom**

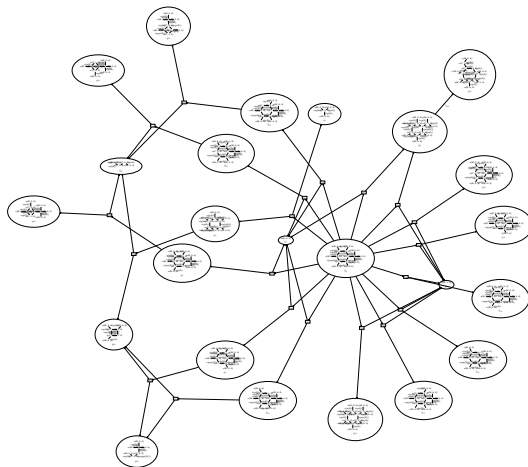
$$\text{BFS} \setminus \{\text{H}, \text{C}\} = \emptyset \times$$

# Predicate – Saturation

**Goal:** Find saturated path  
**Method:**  
Search from **start** to **end**  
following C-C single bonds



# Sample Fragmentation Graph



# Bidirectional Framework

$\mathcal{M}$  molecular structures



molecule graph

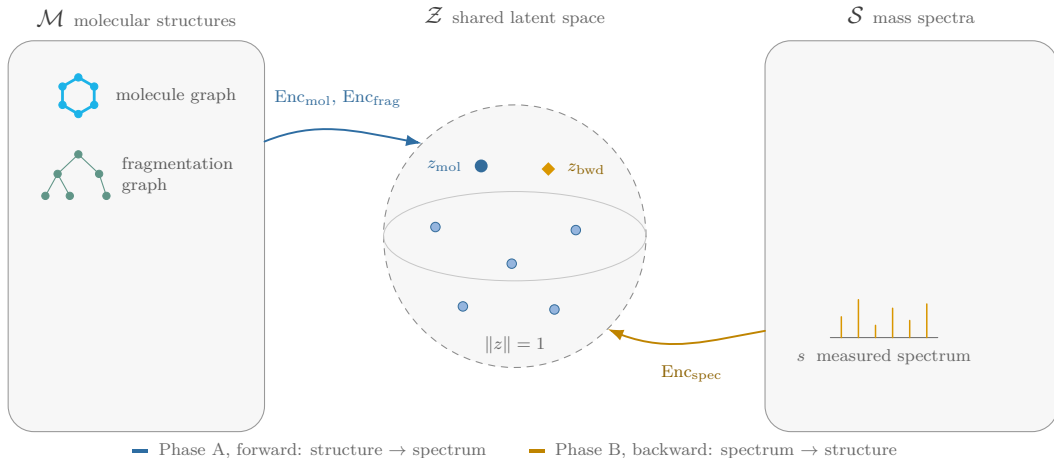


fragmentation  
graph

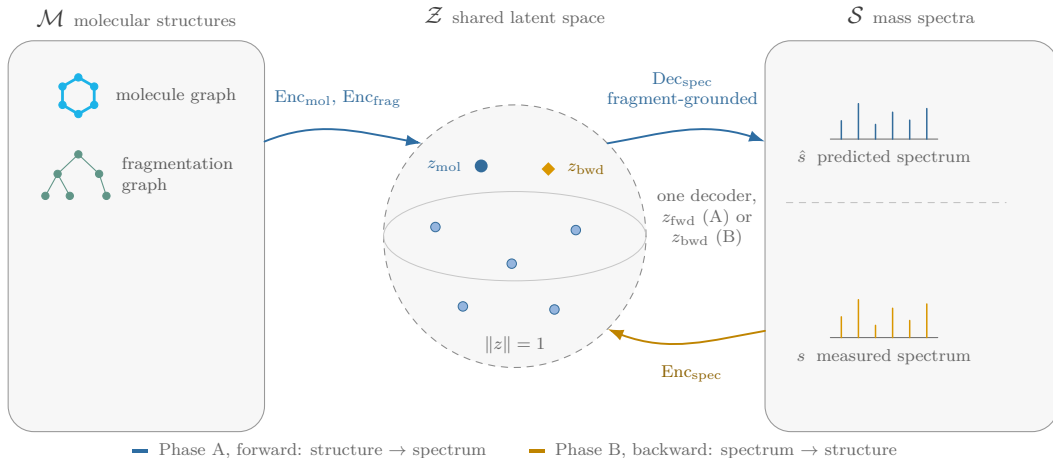
$\mathcal{S}$  mass spectra



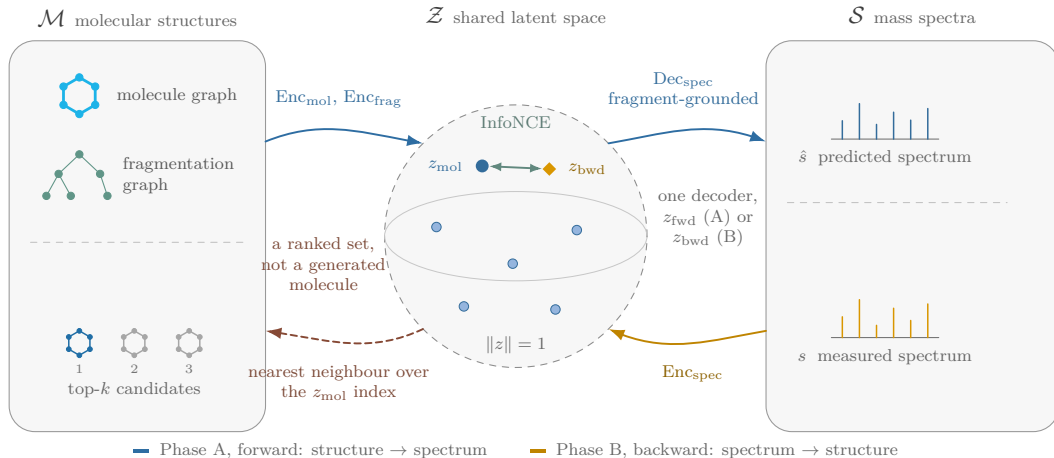
# Bidirectional Framework



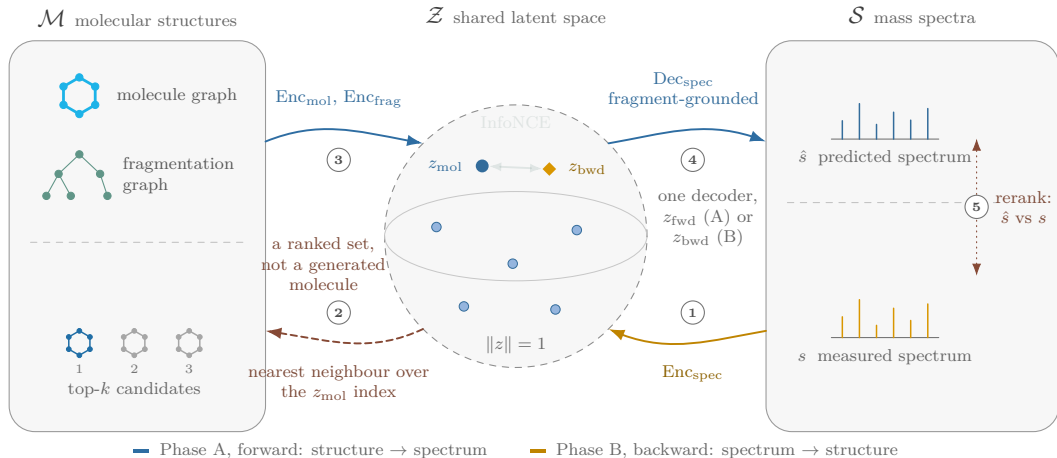
# Bidirectional Framework



# Bidirectional Framework



# Bidirectional Framework – at Query Time



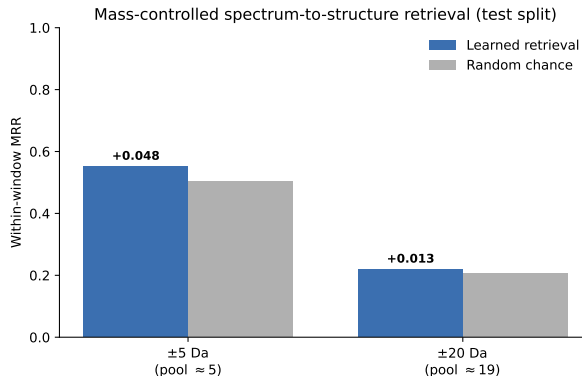
# Results – Mass Sets the Bar

Held-out test split ·  $N = 13$  queries · gallery of 66 EI compounds

| Ranking signal               | MRR   |
|------------------------------|-------|
| True parent mass (oracle)    | 0.923 |
| Spectrum-implied parent mass | 0.462 |
| Mass mask + cosine           | 0.498 |
| Learned embedding            | 0.112 |

⇒ uncontrolled retrieval scores measure **mass agreement**, not chemistry

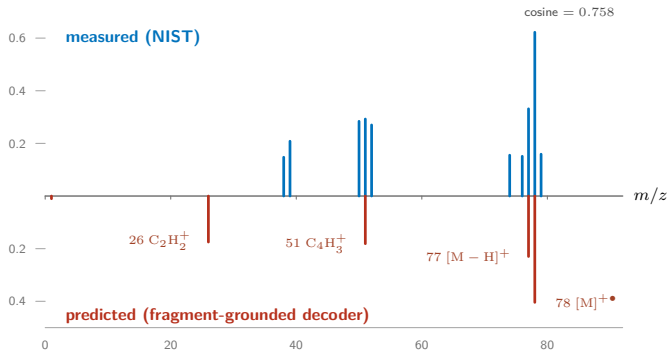
# Results – Mass-Controlled Retrieval



±5 Da: 0.551 vs. chance 0.503 ( $\Delta +0.048$ , mean pool 5.4)  
±20 Da: 0.219 vs. chance 0.206 ( $\Delta +0.013$ , mean pool 18.8)

Inside a window mass is  $\approx$ constant  $\Rightarrow$  the margin is structure the mass cannot explain.

# Results – The Forward Model Is Molecule-Specific



# Conclusions

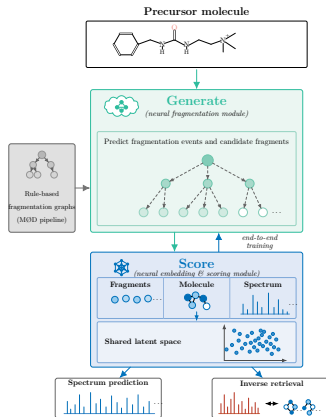
## Contributions

- **Fragment-grounded forward model:** intensities scattered from the molecule's own derived fragments  $\Rightarrow$  molecule-specific *by construction*
- **Mass-controlled evaluation:** separates learned structure from parent-mass filtering

**Supporting work:** bidirectional framework ( $f : \mathcal{M} \rightarrow \mathcal{S}$ ,  $f^{-1} : \mathcal{S} \rightarrow \mathcal{M}$ ) + rule-extension DSL

# Outlook & Literature comparison

## Learn to *Generate* Fragmentation



# References

- [1] Adamo Young, Hannes Röst, and Bo Wang. “Tandem mass spectrum prediction for small molecules using graph transformers”. In: *Nature Machine Intelligence* 6.4 (Apr. 2024), pp. 404–416. ISSN: 2522-5839. DOI: 10.1038/s42256-024-00816-8.
- [2] Yannek Nowatzky et al. “FIORA: Local neighborhood-based prediction of compound mass spectra from single fragmentation events”. In: *Nature Communications* 16.1 (Mar. 2025). ISSN: 2041-1723. DOI: 10.1038/s41467-025-57422-4.
- [3] Mehdi B. Hamaneh et al. “Systematic Assessment of Deep Learning-Based Predictors of Fragmentation Intensity Profiles”. In: *Journal of Proteome Research* 23.6 (May 2024), pp. 1983–1999. ISSN: 1535-3907. DOI: 10.1021/acs.jproteome.3c00857.
- [4] Ivana Blaženović et al. “Software Tools and Approaches for Compound Identification of LC-MS/MS Data in Metabolomics”. In: *Metabolites* 8.2 (May 2018), p. 31. ISSN: 2218-1989. DOI: 10.3390/metabo8020031.
- [5] Jakob L. Andersen et al. “An intermediate level of abstraction for computational systems chemistry”. In: *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 375.2109 (Nov. 2017), p. 20160354. ISSN: 1471-2962. DOI: 10.1098/rsta.2016.0354.
- [6] MØD Developers. *Derivation Graph Strategy (dgStrat) — MØD Documentation*. Accessed 2026-03-24. 2022. URL: <https://imada.sdu.dk/u/daniel/DM840-2022/Material/mod-doc/dgStrat/index.html> (visited on 03/24/2026).
- [7] V8rik. *Toluene Fragmentation*. 2008. URL: <https://upload.wikimedia.org/wikipedia/commons/6/6c/TolueneFragmentation.svg> (visited on 01/20/2026).
- [8] Paginazero, Ronhjones. *Toluene Spectra*. 2017. URL: [https://upload.wikimedia.org/wikipedia/commons/2/2c/Spettro\\_di\\_massa\\_del\\_toluene.PNG](https://upload.wikimedia.org/wikipedia/commons/2/2c/Spettro_di_massa_del_toluene.PNG) (visited on 01/20/2026).
- [9] Fred W McLafferty and František Tureček. *Interpretation von Massenspektren*. ger. Spektrum Lehrbuch. Berlin [u.a.]: Spektrum Akad. Verl., 1995. ISBN: 9783642398483.
- [10] Samuel Goldman, Janet Li, and Connor W. Coley. “Generating Molecular Fragmentation Graphs with Autoregressive Neural Networks”. In: *Analytical Chemistry* 96.8 (Feb. 2024), pp. 3419–3428. ISSN: 1520-6882. DOI: 10.1021/acs.analchem.3c04654.

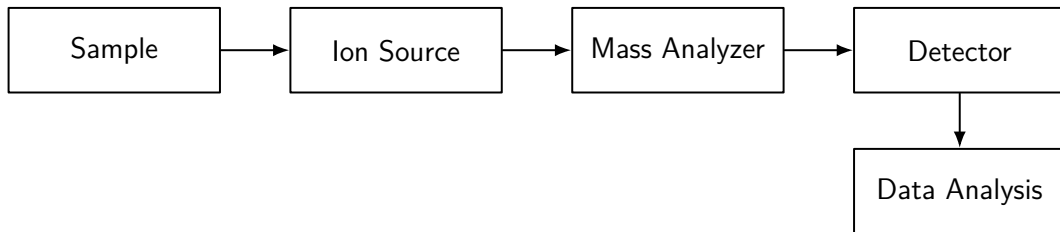
# Discussion

Questions?

## Supplementary Figures

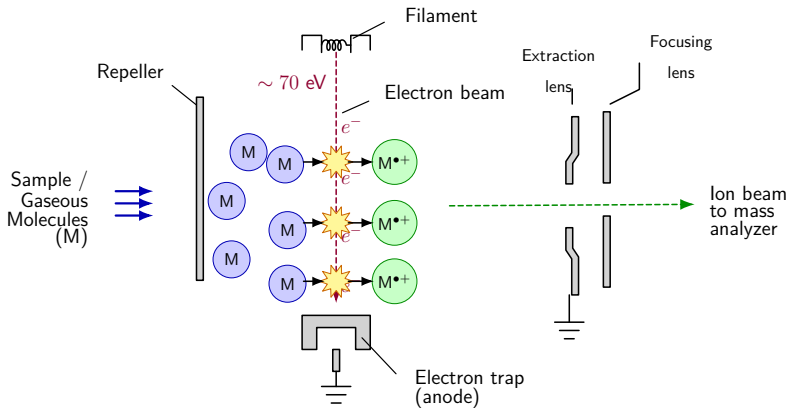
- Instrumentation & ionization methods
- Molecular representation
- Model architecture — training & inference
- The residual MLP block
- Latent-space alignment (InfoNCE)
- Decoder conditioning
- Spectral comparison measures
- Data-generation cost
- What the speedup enabled
- Train / validation / test split

# Mass Spectrometer – Block Diagram



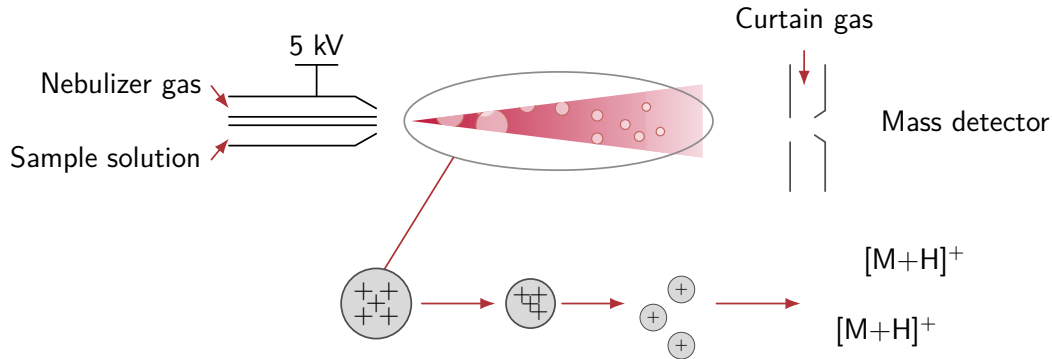
Inlet → ion source → mass analyzer → detector → data analysis.

# Electron Ionization (EI)



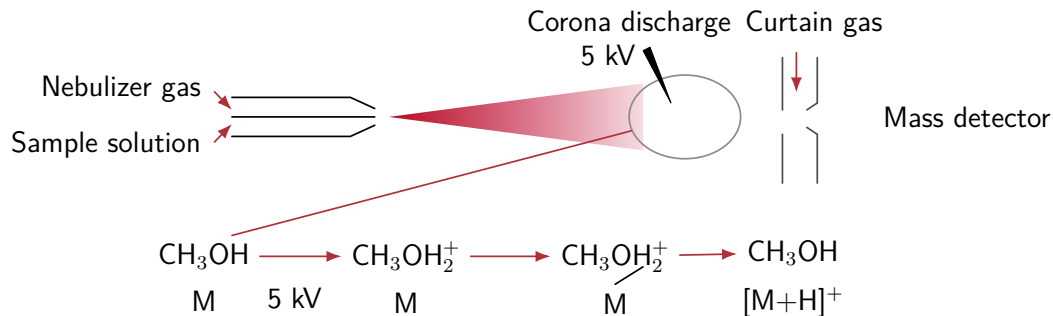
$\sim 70 \text{ eV}$  electrons eject one electron  $\Rightarrow$  radical cation  $[M]^{\bullet+}$ , which then fragments.

# Electrospray Ionization (ESI)



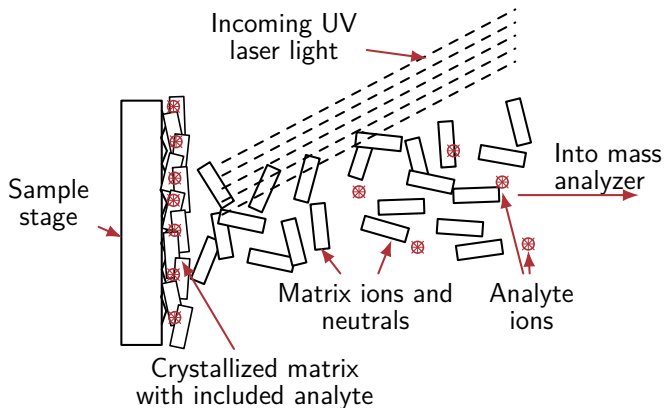
Soft ionization: charged droplets  $\rightarrow$  gas-phase ions, typically  $[M + H]^+$ .

# Atmospheric Pressure Chemical Ionization (APCI)



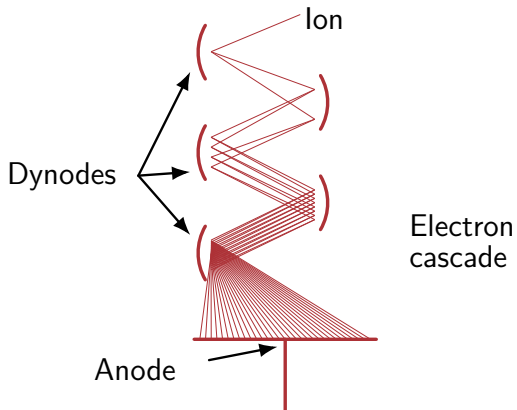
Corona discharge makes reagent ions; proton transfer gives  $[\text{M} + \text{H}]^+$ .

# Matrix-Assisted Laser Desorption/Ionization (MALDI)



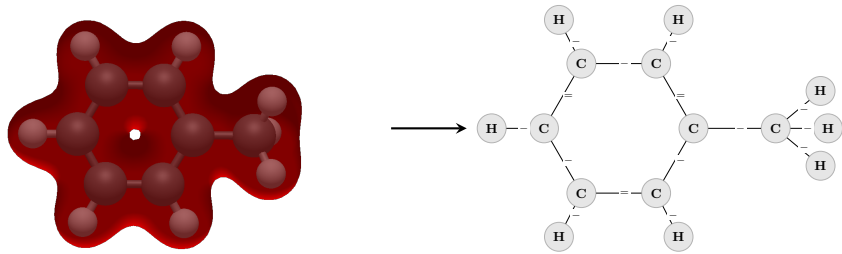
Laser desorption from a matrix; predominantly singly charged ions, suited to large biomolecules.

# Electron Multiplier Detector



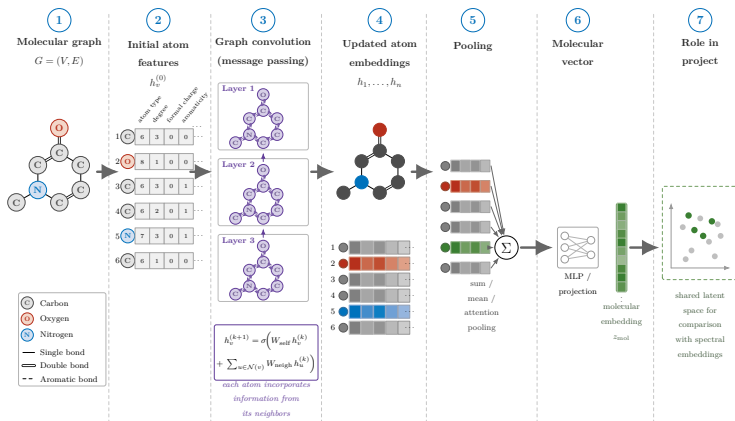
Cascading secondary electrons across dynodes amplify a single ion into a measurable current pulse.

# Molecule as a Graph



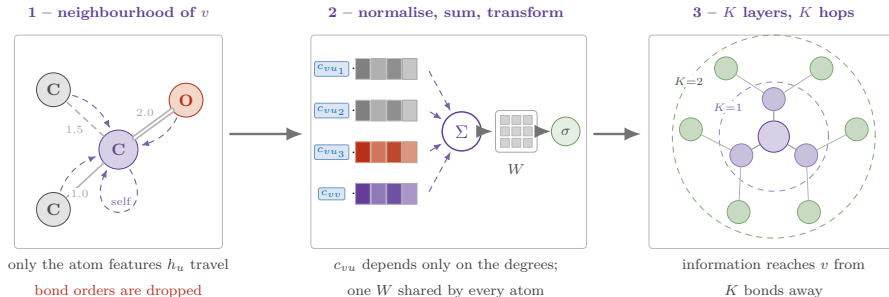
Geometry and electron density are abstracted away; connectivity is kept.

# Graph-Convolutional Molecular Encoding



Atom features  $\rightarrow$  message passing  $\rightarrow$  pooling  $\rightarrow z_{mol}$  in the shared latent space.

# What a Graph Convolution Does

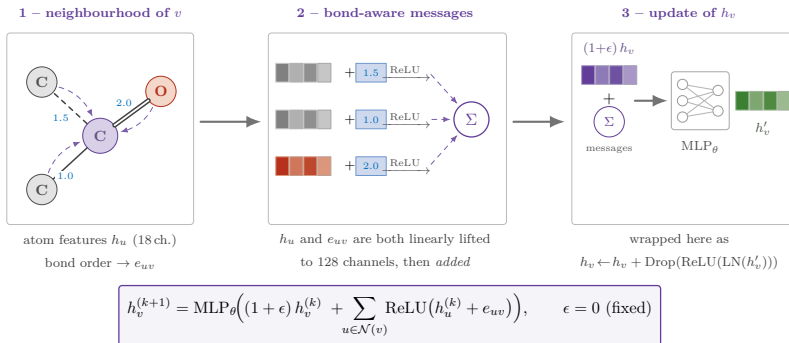


$$h_v^{(k+1)} = \sigma \left( W^{(k)} \sum_{u \in \mathcal{N}(v) \cup \{v\}} c_{vu} h_u^{(k)} \right), \quad c_{vu} = \frac{1}{\sqrt{\hat{d}_v \hat{d}_u}}$$

no  $e_{uv}$  in the sum  $\Rightarrow$  single, double and aromatic bonds send the same message

Each atom is replaced by a degree-weighted average of its neighbours;  $K$  layers  $\Rightarrow K$ -hop context.

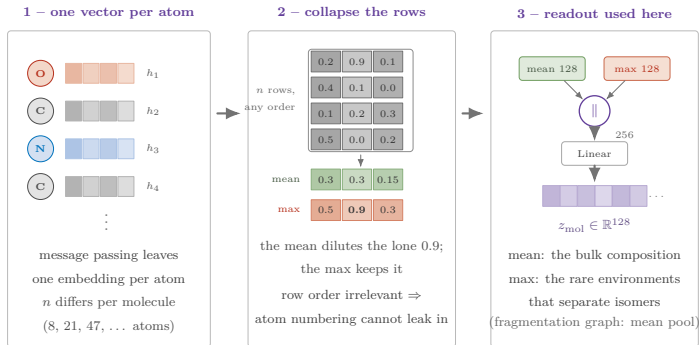
# Why GINEConv – Bond Orders Enter the Message



A GCN averages neighbours and *drops* the bond, so what survives is composition — i.e. mass. GINE adds the bond embedding to every message and *sums* (1-WL expressive)  $\Rightarrow$  constitutional isomers stay distinguishable.

Here:  $3 \times 128$  layers, LayerNorm + residual, mean  $\parallel$  max readout  $\rightarrow z_{\text{mol}} \in \mathbb{R}^{256}$ .

# Pooling – $n$ Atom Vectors $\rightarrow$ One Molecule Vector



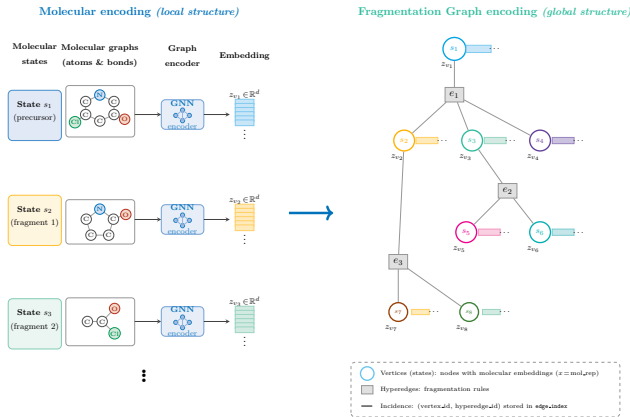
$$z_{\text{mol}} = W \left[ \text{mean}_{v \in V} h_v \parallel \max_{v \in V} h_v \right] \in \mathbb{R}^{128}, \quad \text{unchanged by any permutation of } V$$

# Two-Level Representation – Overview



Local graph encoders for atom/bond chemistry, global hypergraph encoders for multi-step fragmentation.

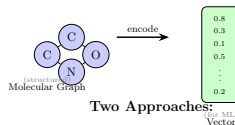
# Two-Level Molecular Latent Representation



Stage 1: state  $\rightarrow$  embedding. Stage 2: embeddings become vertex features of the fragmentation hypergraph.

# The Graph Representation Challenge

## Graph Representation Challenge



### Two Approaches:

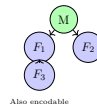


- ✓ Better performance
- ✗ Not invertible



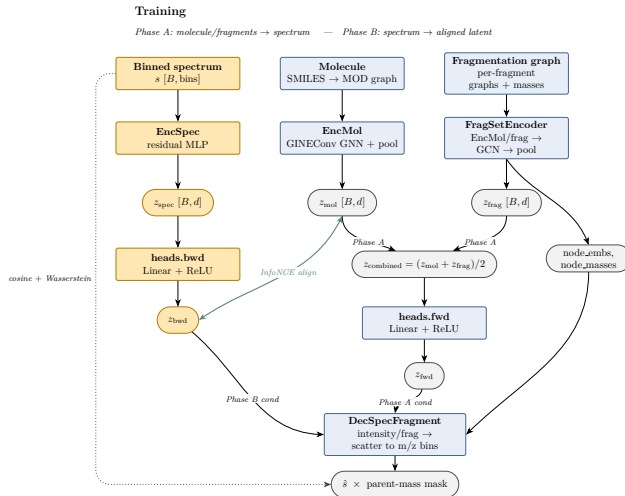
- ✓ Structured info
- ✓ For fragments too

### Fragmentation Trees



Fingerprints score well but are not (easily) invertible — which matters for the inverse direction.

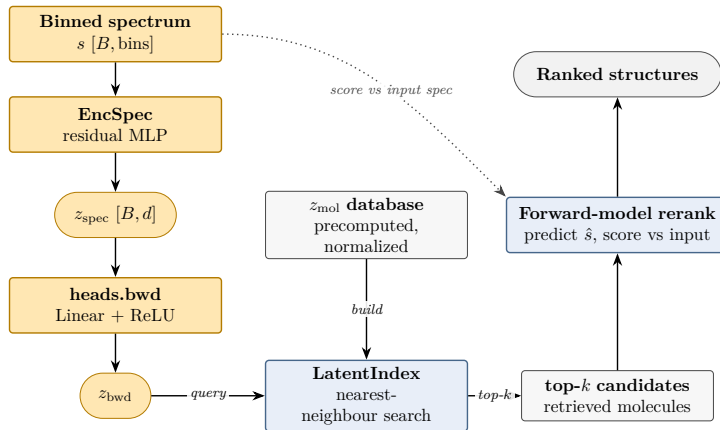
# Architecture – Training



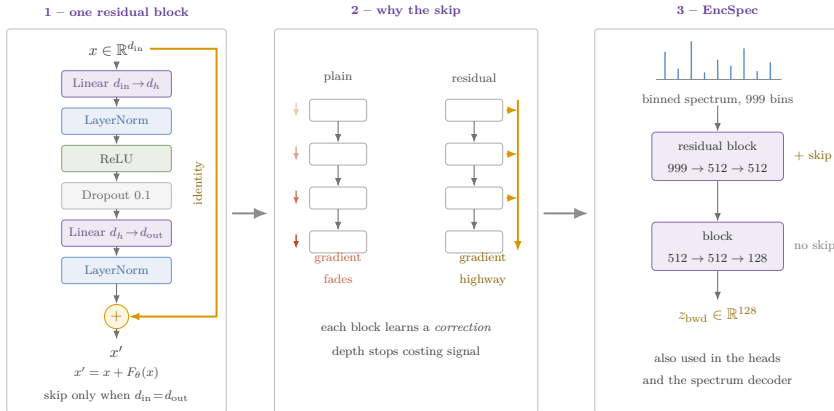
# Architecture – Inference

## Inference / retrieval

Query with  $z_{\text{bwd}}$  into the molecule-latent index, then rerank with the forward model

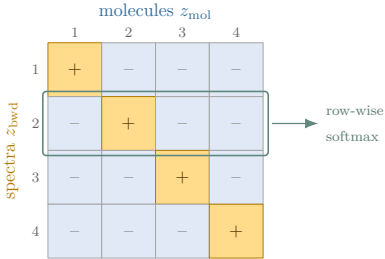
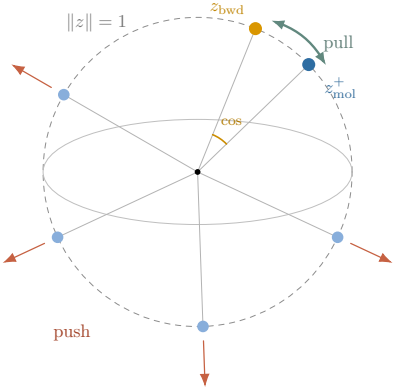


# The Residual MLP Block



$x' = x + F_{\theta}(x)$  — the block learns a *correction*, not the whole map.

# Phase B – InfoNCE Alignment

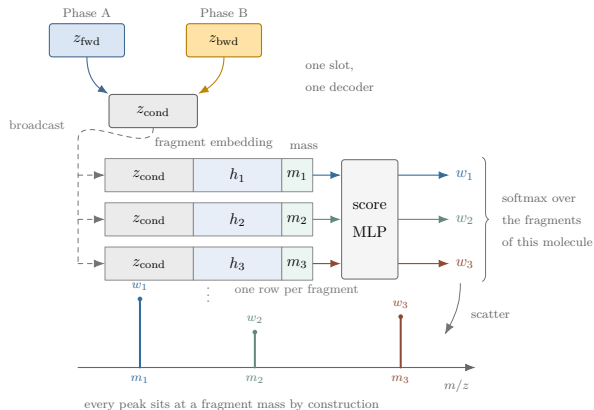


$$\mathcal{L}_{NCE} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(z_{bwd}^i \cdot z_{mol}^i / T)}{\sum_{j=1}^B \exp(z_{bwd}^i \cdot z_{mol}^j / T)}$$

$T = 0.07$     batches sorted by parent mass  
 $\Rightarrow$  the negatives are near-isobaric

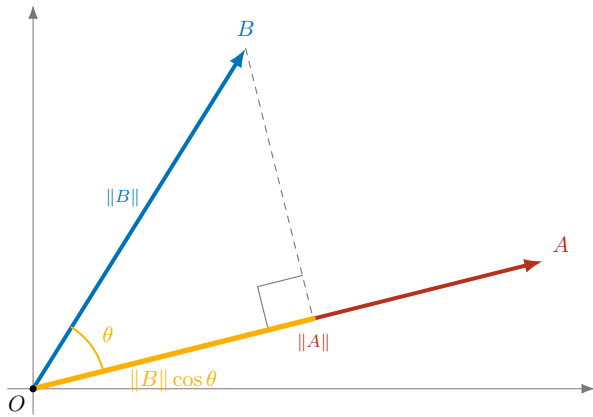
One shared space    · diagonal = positive, rest of the batch = negatives

# Conditioning the Decoder



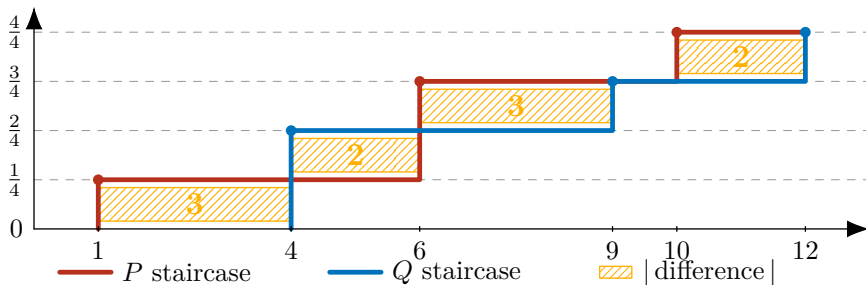
One decoder, one slot: Phase A puts  $z_{\text{fwd}}$  in it, Phase B  $z_{\text{bwd}}$ .

# Cosine Similarity – Geometry



$\cos \theta = (A \cdot B) / (\|A\| \|B\|)$  — direction only, hence scale-invariant.

# Wasserstein Distance – Illustration

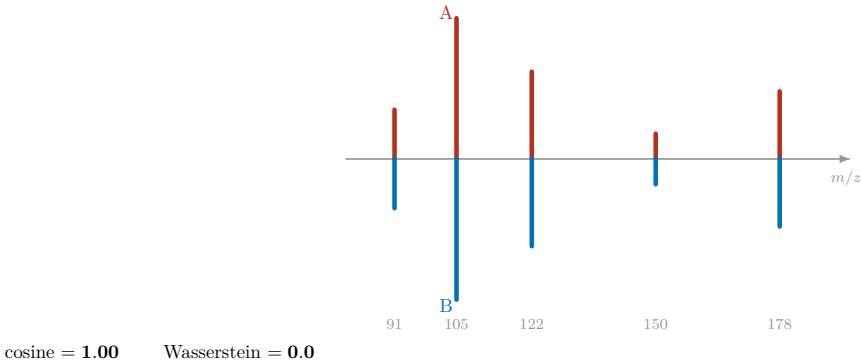


Sorted vectors as cumulative staircases; the distance is the shaded transport area, normalized by  $n$ .



# Cosine vs. Wasserstein – (a) Aligned Spectra

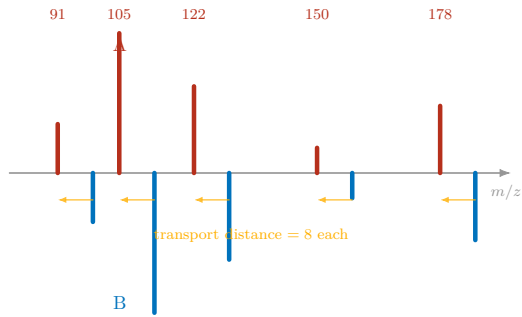
(a) Aligned spectra — identical,  $\theta = 0$



Every B peak meets its A counterpart: cosine = 1 ( $\theta = 0$ ), Wasserstein = 0 (no mass to move).

# Cosine vs. Wasserstein – (b) B Shifted +8 Da

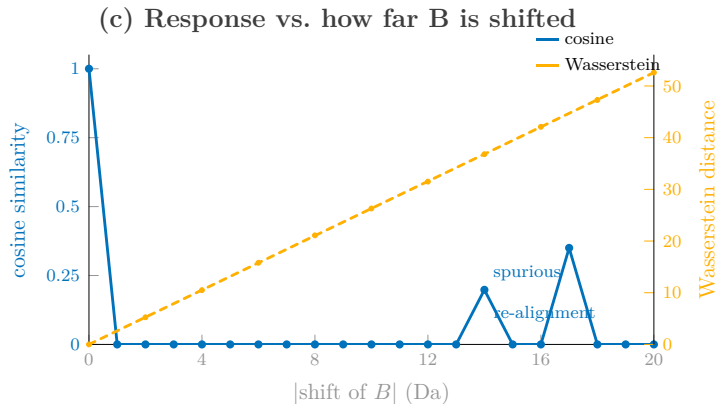
(b) B shifted +8 Da — peaks miss,  $\theta \rightarrow 90^\circ$



cosine = **0.00** (no shared peaks)      Wasserstein = **21.1**

Cosine counts only *shared* positions  $\Rightarrow$  collapses to 0; Wasserstein transports the mass back and stays finite (21.1).

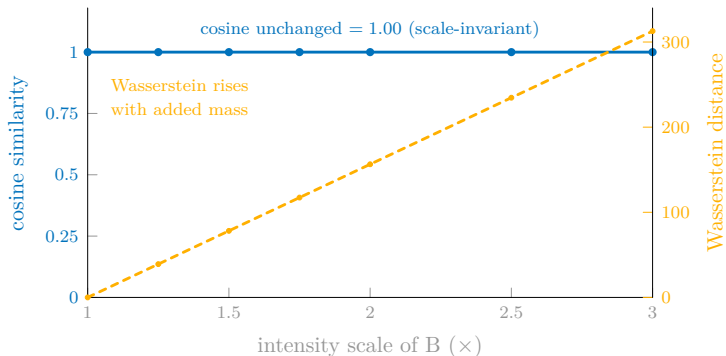
# Cosine vs. Wasserstein – (c) Response vs. Shift



Cosine drops to zero at once, with spurious spikes at  $\sim 14$  and  $17$  Da; Wasserstein rises linearly with displacement.

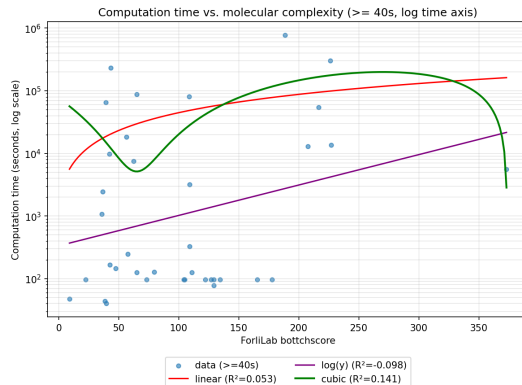
# Cosine vs. Wasserstein – (d) Response vs. Scaling

(d) Response vs. B's peak intensity (positions aligned)



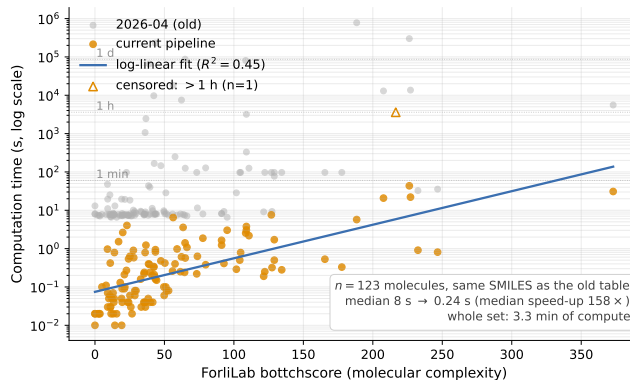
Cosine unchanged at 1.00 (direction, not magnitude); un-normalized Wasserstein grows with the added intensity.

# Fragmentation Computation Times



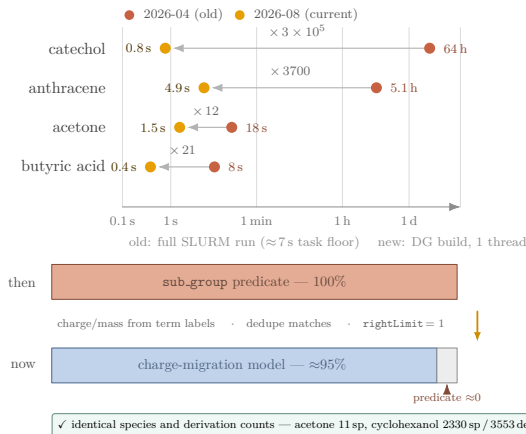
Time vs. molecular complexity (trivial expansions  $< 40$  s excluded): positive trend, but huge variance at equal complexity.

# Fragmentation Computation Times – Re-measured



Same molecules, current code · the trend survives, the level drops

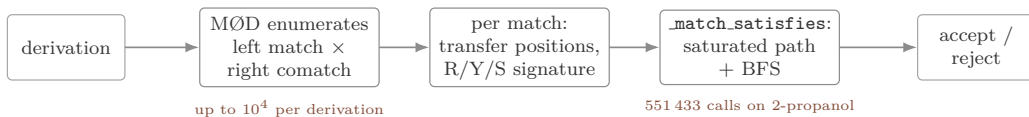
# Fragmentation Cost – Then vs Now



Same fragments, more chemistry · the bottleneck moved

# Fragmentation Cost – What Moved

The hot path, 2026



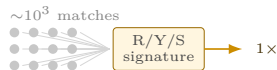
What moved it

① Read, don't rebuild

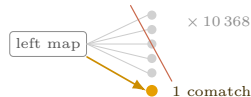
~~term → SMILES → molecule →  $q, m$~~

term label →  $q, m$

② Memoise the traversal



③ Cap the comatch fan



# Where the Data-Generation Time Goes

## MØD data generation — sampled call tree (folic\_acid, forward-only, 1 thread)

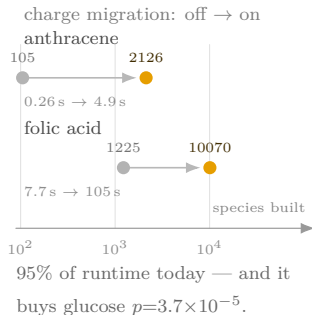
width = wall time · 11.662 stack samples over 117.9 s of build (imports pre-warmed) · 86% of samples are inside MØD's C++ matching loop with no Python frame below



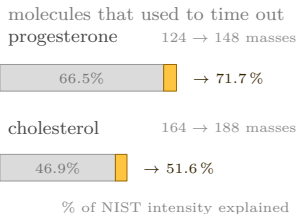
Sampled call tree (folic acid, forward-only, 1 thread): 86% of samples sit inside MØD's C++ matching loop.

# What the Speedup Bought

## ① New chemistry



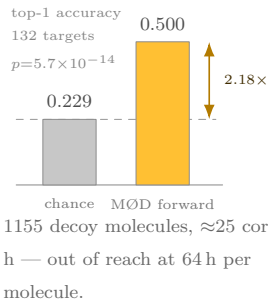
## ② The hard tail



✓ aldosterone: killed at 12 h30  
→ 1 h14. A timeout writes no  
dump.

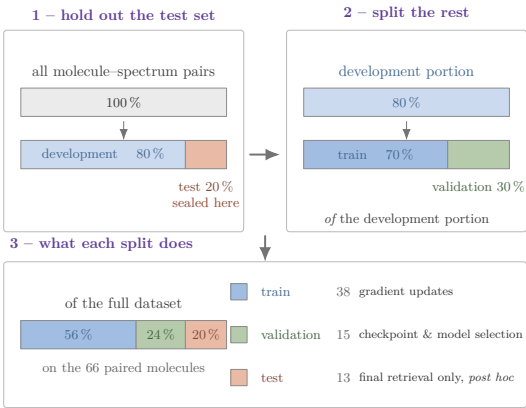
## ③ 1340 decoys

same-formula discrimination



Faster is not the point · what the compute was spent on is

# Train / Validation / Test – Two-Stage Split



shuffle=False — deterministic, reproducible by construction

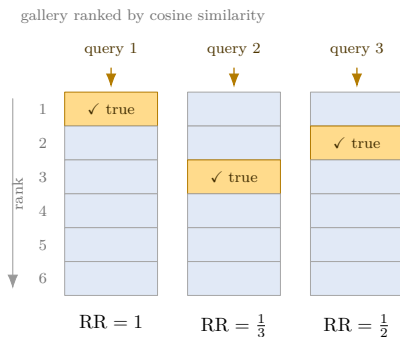
$$56 / 24 / 20 \% \Rightarrow 38 / 15 / 13 \text{ molecules}$$

## Splits, Gallery and Provenance

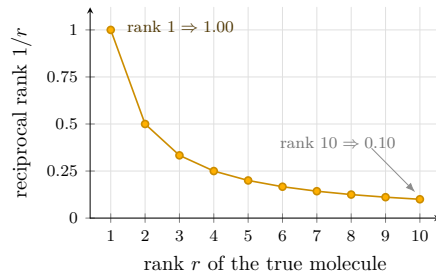
|                                        |              |
|----------------------------------------|--------------|
| Molecules with paired dumps + spectra  | 66           |
| train / validation / test              | 38 / 15 / 13 |
| Retrieval gallery (train + val + test) | 66           |
| Mass-controlled queries (test)         | 13           |
| Training runs behind the numbers       | 1            |
| Spectrum bins (1–1000 Da, 1 Da)        | 999          |

All reported numbers come from one training run (`ml_checkpoint.pt`, 2026-07-06) — no seed averaging, no model selection on the test split.

# Mean Reciprocal Rank (MRR)



$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{r_i} = \frac{1 + \frac{1}{3} + \frac{1}{2}}{3} = 0.61$$



Only the rank of the *true* molecule counts · the top of the list dominates

# Analytic Chance Level in a Mass Window

A query restricted to  $\pm W$  Da leaves a pool of  $m$  candidates, exactly one of which is correct. Ranking that pool at random puts the correct molecule at rank  $r$  with probability  $1/m$ , so

$$\text{MRR}_{\text{chance}}(m) = \mathbb{E}\left[\frac{1}{r}\right] = \frac{1}{m} \sum_{r=1}^m \frac{1}{r} = \frac{H_m}{m}, \quad \text{chance} = \frac{1}{N} \sum_{i=1}^N \frac{H_{m_i}}{m_i}$$

$H_m$ : the  $m$ -th harmonic number;  $m_i$ : pool size of query  $i$ .

| $m$     | 1     | 2     | 3     | 5     | 10    | 20    |
|---------|-------|-------|-------|-------|-------|-------|
| $H_m/m$ | 1.000 | 0.750 | 0.611 | 0.457 | 0.293 | 0.180 |

The average is taken *per query*, not at the mean pool size:  $H_m/m$  is convex, so plugging in  $\bar{m} = 5.4$  would give  $\approx 0.44$  instead of 0.503 — the per-query bar is the **harder** one.

## Honest assessment

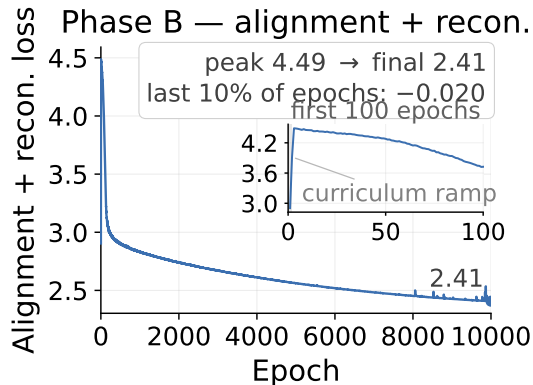
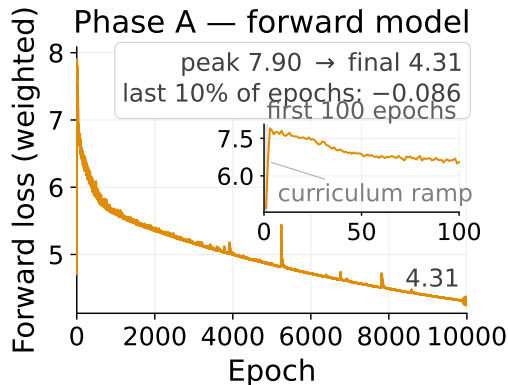
- Beyond-mass margin is *small*: +0.048 / +0.013 MRR on 13 queries, single run
- Uncontrolled metrics — reranking included — are mass filtering, not evidence
- Fragment branch buys molecule-specificity, not a retrieval win

# Training Hyperparameters

|                                   |                                             |
|-----------------------------------|---------------------------------------------|
| Optimizer                         | Adam (one per phase)                        |
| Learning rate / weight decay      | $7.1 \times 10^{-5}$ / $1.4 \times 10^{-4}$ |
| Batch size                        | 64                                          |
| Epochs (Phase A + Phase B)        | 10 000 + 10 000                             |
| Shared latent dimension $d$       | 256                                         |
| GINEConv layers / width           | 3 / 128                                     |
| Fragmentation-graph GCN           | 2× GCNConv, mean pool                       |
| Dropout / normalization           | 0.43 / BatchNorm                            |
| Cosine / Wasserstein weight       | 1.59 / 1.17                                 |
| Cycle (fwd / spec) weight         | 0.77 / 0.089                                |
| Align weight / temperature $\tau$ | 0.30 / 0.07                                 |
| Curriculum start / increment      | 0.73 / 0.17                                 |

Tuned entries: best of a 20-trial Optuna sweep; checkpoint selection = final epoch.

## Training Loss – Both Phases



Training loss only · no per-epoch validation logged · checkpoint = final epoch

## Model Size – Where the Parameters Are

| Module                                                                             | Parameters       | Share |
|------------------------------------------------------------------------------------|------------------|-------|
| Molecule encoder ENCMOL ( $3 \times$ GINEConv, $w=128$ )                           | 168 320          | 9.9%  |
| Fragmentation-graph GCN ( $2 \times$ GCNConv)                                      | 33 152           | 1.9%  |
| Spectrum encoder ENCSPEC ( $999 \rightarrow 512 \rightarrow 512 \rightarrow 256$ ) | 1 170 688        | 68.8% |
| Fragment-grounded decoder DECSPECFRAGMENT                                          | 198 657          | 11.7% |
| Task heads (fwd / bwd + loss $\log \sigma^2$ )                                     | 131 586          | 7.7%  |
| <b>Total (trainable)</b>                                                           | <b>1 702 403</b> |       |

The single  $999 \times 512$  input matrix of the spectrum encoder is 30% of the whole model.